VISIBLE SPECTRUM: WHEN ADAPTIVE OPTIMIZERS CHANGE SCALING LAWS

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ABSTRACT

Scaling laws are usually stated as properties of model families, datasets, and compute budgets, while the optimizer is often treated as affecting only constants. We show that, even in linear regression, this is false in a precise and diagnosable way. In power-law Gaussian regression, coordinatewise adaptive methods such as RMSProp, Adam, and AdamW change scaling exponents only when their coordinatewise second moments expose spectral information about the data covariance. We formalize this through a *visible spectrum* exponent $\theta \in [0,1]$: if the adaptive preconditioner is spectrally comparable to $(\Sigma^\theta + \rho I)^{-1/2}$, then the effective covariance eigenvalues scale as

$$\widetilde{\lambda}_i \asymp \lambda_i(\lambda_i^\theta + \rho)^{-1/2}, \qquad q_{\text{eff}} = \theta/2.$$

Consequently, for $\lambda_i \asymp i^{-a}$ the learned-mode count changes from the SGD rate $n^{1/a}$ to $n^{1/[a(1-\theta/2)]}$ before the damping knee. Under a source condition $s_i \asymp i^{-b}$, the excess risk obeys

$$R_M - \sigma^2 \asymp M^{1-b} + K_{\rho,\theta}(n)^{1-b} + \frac{K_{\rho,\theta}(n)}{N_{\text{eff}}},$$

with corresponding compute-optimal exponent $(b-1)/(\max\{a(1-\theta/2), b\}+1)$. AdamW weight decay imposes a learned-mode cap; to preserve the no-decay compute law, the effective decay must decrease with compute. Experiments in deterministic filters, stochastic training, coordinate rotations, sketched/random-feature models, and compute-optimal sweeps support the theory. The results suggest that adaptive optimizers change scaling laws through *visible spectral information*, not through adaptivity alone.

1 INTRODUCTION

Empirical neural scaling laws describe predictable power-law improvements as model size, data, and compute increase. Recent theoretical work has shown that related scaling laws already arise in infinite-dimensional linear regression trained by one-pass SGD under power-law covariance spectra (Lin et al., 2024). In those models, the reducible risk decomposes into approximation and optimization terms, and the variance term is dominated by SGD's implicit regularization.

This paper asks a complementary question:

> *When do adaptive optimizers change scaling exponents rather than merely constants?*

The answer we propose is: *when coordinatewise second moments expose the spectrum.* Coordinatewise adaptive methods do not see covariance eigenvalues directly. They see the diagonal of the covariance in the optimizer coordinate system. If optimizer coordinates are aligned with covariance eigendirections, RMSProp/Adam's second moment is proportional to λ_i and the method behaves like damped spectral preconditioning with exponent $q = 1/2$. If coordinates are globally mixed so that all coordinate variances are flat, the same adaptive rule becomes scalar up to constants and cannot change the spectral exponent. Intermediate feature systems produce intermediate visible exponents.

Main mechanism. Let $z \sim \mathcal{N}(0, \Sigma)$ be the trainable-coordinate feature vector. For squared loss and error vector $e = w - w^\star$, the gradient is

$$g = (\langle z, e\rangle - \xi)z.$$

A Gaussian fourth-moment calculation gives

$$\mathbb{E}[g_j^2 \mid e] = (\sigma^2 + \|e\|_\Sigma^2)\Sigma_{jj} + 2(\Sigma e)_j^2 \asymp (\sigma^2 + \|e\|_\Sigma^2)\Sigma_{jj}.$$

Thus RMSProp/Adam tracks coordinate variances Σ_{jj}, not eigenvalues. The preconditioned covariance is

$$\widetilde{\Sigma} = P^{1/2}\Sigma P^{1/2}, \qquad P = \mathrm{diag}((\Sigma_{jj} + \rho)^{-1/2}).$$

If P is spectrally comparable to $(\Sigma^\theta + \rho I)^{-1/2}$, then $q_{\text{eff}} = \theta/2$. The entire scaling law follows from this visible spectral exponent.

Contributions.

1. We derive a visible-spectrum theorem for coordinatewise adaptive optimizers. The theorem interpolates between scalar preconditioning ($\theta = 0$) and Adam/RMSProp spectral preconditioning ($\theta = 1$).
2. We prove sharp risk and compute-optimal scaling laws in the visible-spectrum model, including the phase transition where increasing visibility no longer improves the compute exponent.
3. We analyze Adam components separately: RMSProp-like second moments determine the spectral exponent; first-moment momentum changes constants/stability; AdamW weight decay adds a learned-mode cap and must decay with compute to preserve scaling.
4. We prove alignment results showing that eigenbasis and band-limited feature maps expose the spectrum, while flat/Haar/global Gaussian sketches hide it.
5. We validate the mechanism in deterministic filter experiments, stochastic training, learning-rate sweeps, compute-optimal sweeps, and random-feature/sketched-coordinate experiments.

2 Setup

Let $x \sim \mathcal{N}(0, H)$ and $y = \langle x, w^\star\rangle + \xi$, with $\mathbb{E}\xi^2 = \sigma^2$. We study a trainable feature representation $z = Sx$ with covariance

$$\Sigma = \mathbb{E}[zz^\top] = SHS^\top.$$

The eigenvalues of Σ are denoted $\lambda_i(\Sigma)$ and are assumed to satisfy

Assumption 1 (Power-law spectrum)**.** *For some $a > 1$,*

$$\lambda_i(\Sigma) \asymp i^{-a}.$$

For spectral statements we write $\Sigma v_i = \lambda_i v_i$. The source energies are

$$s_i = \lambda_i \langle v_i, w^\star\rangle^2.$$

Assumption 2 (Source condition)**.** *For some $b > 1$,*

$$s_i \asymp i^{-b},$$

possibly in the transformed basis when the preconditioner and Σ do not commute. In aligned and invariant band-limited cases, this transformed source condition follows from source-mass preservation (Appendix A.1).

We write N for the number of samples/updates and $N_{\text{eff}} = N/\log N$ for the effective one-pass horizon, following the logarithmic convention in prior work. The model dimension is M.

3 VISIBLE SPECTRUM OF COORDINATEWISE ADAPTIVE METHODS

3.1 GRADIENT SECOND MOMENTS REVEAL COORDINATE VARIANCES

Lemma 1 (Coordinate second moments). *Let $z \sim \mathcal{N}(0, \Sigma)$, $y = \langle z, w^\star \rangle + \xi$, and $e = w - w^\star$. For the squared-loss gradient*

$$g = (\langle z, e \rangle - \xi) z,$$

we have for each coordinate j,

$$\mathbb{E}[g_j^2 \mid e] = c_e \Sigma_{jj} + 2(\Sigma e)_j^2, \qquad c_e = \sigma^2 + \|e\|_\Sigma^2.$$

Consequently,

$$c_e \Sigma_{jj} \le \mathbb{E}[g_j^2 \mid e] \le 3 c_e \Sigma_{jj}.$$

Proof. Let $U = \langle z, e \rangle - \xi$ and $Z_j = z_j$. Conditionally on e, (U, Z_j) is centered Gaussian with

$$\mathbb{E}U^2 = c_e, \qquad \mathbb{E}Z_j^2 = \Sigma_{jj}, \qquad \mathbb{E}[UZ_j] = (\Sigma e)_j.$$

Isserlis' identity gives

$$\mathbb{E}[U^2 Z_j^2] = \mathbb{E}U^2 \mathbb{E}Z_j^2 + 2\mathbb{E}[UZ_j]^2 = c_e \Sigma_{jj} + 2(\Sigma e)_j^2.$$

The upper bound follows from

$$(\Sigma e)_j^2 \le (e^\top \Sigma e)\Sigma_{jj} \le c_e \Sigma_{jj}.$$

□

Thus RMSProp/Adam second moments track $d_j = \Sigma_{jj}$ up to scalar residual factors. The scalar affects the learning rate and damping but not the spectral exponent.

3.2 SPECTRAL COMPARABILITY AND THE VISIBLE EXPONENT

Definition 1 (Visible spectral exponent). *A coordinatewise adaptive preconditioner P has visible exponent $\theta \in [0, 1]$ at damping ρ if, for constants $0 < c < C < \infty$ independent of scale,*

$$c(\Sigma^\theta + \rho I)^{-1/2} \preceq P \preceq C(\Sigma^\theta + \rho I)^{-1/2}. \tag{1}$$

Theorem 1 (Visible-spectrum theorem). *Assume (1). Let $\widetilde{\Sigma} = P^{1/2} \Sigma P^{1/2}$. Then*

$$\lambda_i(\widetilde{\Sigma}) \asymp \lambda_i(\Sigma)\big(\lambda_i(\Sigma)^\theta + \rho\big)^{-1/2}.$$

In particular, if $\lambda_i(\Sigma) \asymp i^{-a}$ and the damping is inactive on the fitted range, then

$$\lambda_i(\widetilde{\Sigma}) \asymp i^{-a(1-\theta/2)}.$$

Thus

$$q_{\text{eff}} = \theta/2.$$

Proof. The nonzero eigenvalues of $P^{1/2}\Sigma P^{1/2}$ equal those of $\Sigma^{1/2} P \Sigma^{1/2}$. Applying the Loewner comparison and using that Σ commutes with $(\Sigma^\theta + \rho I)^{-1/2}$ gives

$$c\,\Sigma(\Sigma^\theta + \rho I)^{-1/2} \preceq \Sigma^{1/2} P \Sigma^{1/2} \preceq C\,\Sigma(\Sigma^\theta + \rho I)^{-1/2}.$$

The eigenvalues of the outer terms are exactly

$$\lambda_i(\Sigma)(\lambda_i(\Sigma)^\theta + \rho)^{-1/2}.$$

The min–max principle proves the result. □

Corollary 1 (Learned-mode count). *Let $n = N_{\text{eff}}\gamma$ be the effective optimization horizon. Define*

$$K_{\rho,\theta}(n) = \#\{i : \lambda_i(\widetilde{\Sigma}) \gtrsim n^{-1}\}.$$

For $\theta > 0$,

$$K_{\rho,\theta}(n) \asymp \begin{cases} n^{1/[a(1-\theta/2)]}, & n \lesssim \rho^{-(1/\theta - 1/2)}, \\ \rho^{-1/(2a)} n^{1/a}, & n \gtrsim \rho^{-(1/\theta-1/2)}. \end{cases}$$

For $\theta = 0$, $K_{\rho,0}(n) \asymp n^{1/a}$.

Interpretation. Aligned or band-limited coordinates expose $\theta \approx 1$ and therefore $q_{\text{eff}} \approx 1/2$. Flat, Haar-mixed, or isotropic Gaussian-sketch coordinates expose $\theta \approx 0$ and therefore $q_{\text{eff}} \approx 0$.

4 Risk and compute scaling

The visible-spectrum theorem reduces the analysis to SGD on a transformed covariance. Under Assumptions 1–2, with effective exponent

$$\alpha = a(1 - \theta/2),$$

we obtain the following filter law.

Theorem 2 (Visible-spectrum risk law)**.** *In the clean source range $1 < b < \alpha+1$, one-pass training with an adaptive preconditioner satisfying Theorem 1 obeys, up to logarithmic and constant factors,*

$$\mathbb{E}R_M(w_N) - \sigma^2 \asymp M^{1-b} + K_{\rho,\theta}(n)^{1-b} + \frac{\min\{M, K_{\rho,\theta}(n)\}}{N_{\text{eff}}}.$$

A more exact statement replaces the second and third terms by the spectral filters

$$B(n, M) = \sum_{i \le M} \frac{\widetilde{s}_i}{1 + n\widetilde{\mu}_i}, \qquad V(n, M) = \frac{1}{N_{\text{eff}}} \sum_{i \le M} \min\{1, n^2 \widetilde{\mu}_i^2\}.$$

The simplified learned-count form follows by integral comparison for two-slope spectra.

4.1 Compute-optimal allocation

Let compute be $C \asymp MN$. In the pre-damping regime, the simplified risk proxy is

$$R(M, N, K) - \sigma^2 \asymp M^{1-b} + K^{1-b} + K/N,$$

subject to

$$K \le N^{1/\alpha}.$$

Theorem 3 (Compute-optimal visible scaling)**.** *Let*

$$m = \max\{\alpha, b\}, \qquad \alpha = a(1 - \theta/2).$$

Then

$$M_\star(C) \asymp C^{1/(m+1)}, \qquad N_\star(C) \asymp C^{m/(m+1)},$$
$$K_\star(C) \asymp C^{1/(m+1)},$$

and

$$R_\star(C) - \sigma^2 \asymp C^{-(b-1)/(m+1)}.$$

Equivalently, the compute-risk exponent is

$$\beta(\theta) = \frac{b-1}{\max\{a(1-\theta/2), b\} + 1}.$$

Corollary 2 (Visibility phase transition)**.** *Define*

$$\theta_c = 2\left(1 - \frac{b}{a}\right).$$

When $0 < \theta_c < 1$,

$$\beta(\theta) = \begin{cases} \dfrac{b-1}{a(1-\theta/2)+1}, & 0 \le \theta \le \theta_c, \\ \dfrac{b-1}{b+1}, & \theta \ge \theta_c. \end{cases}$$

Thus visibility improves the compute exponent until the variance-limited phase, then saturates.

4.2 AdamW weight decay

AdamW decoupled weight decay adds the update

$$w_{t+1} = (1-\gamma\delta)w_t - \gamma P_t m_t.$$

For a fixed effective spectrum μ_i, the population coordinate error contains the shrinkage floor

$$\frac{\delta}{\mu_i + \delta}.$$

Thus learned modes must satisfy $\mu_i \gtrsim \max\{n^{-1}, \delta\}$.

If $\delta(C) = C^{-s}$, the weight-decay cap is

$$K \le \delta(C)^{-1/\alpha} = C^{s/\alpha}.$$

Theorem 4 (Compute-dependent AdamW decay)**.** *Let $m = \max\{\alpha, b\}$ and $\delta(C) = C^{-s}$. The compute-risk exponent under the weight-decay cap is*

$$\beta_{\mathrm{wd}}(\theta, s) = \min\left\{\frac{(b-1)s}{\alpha}, \frac{b-1}{m+1}\right\}.$$

The no-decay compute law is recovered iff

$$s \ge \frac{\alpha}{m+1}.$$

Fixed weight decay corresponds to $s = 0$ and creates an asymptotic floor.

5 Experiments

We summarize the empirical evidence. The experiments are synthetic by design: each isolates one theoretical mechanism.

5.1 Deterministic filters and coordinate visibility

Deterministic filter sweeps verify the learned-count predictions for SGD, Adam/RMSProp-like spectra, and AdamW weight decay. For example, when $a = 1.5, b = 1.4$, Adam/RMSProp count slopes match the predicted 1.333; AdamW slopes follow the predicted cap when weight decay is active. Coordinate-visibility sweeps show aligned coordinates have effective slope $-a/2$, while flat/Haar coordinates retain slope $-a$.

5.2 Stochastic training and learning-rate sweeps

Actual mini-batch training confirms that RMSProp/Adam/AdamW second moments learn the predicted q_{eff} in aligned and band-limited coordinates, while flat/Haar coordinates have $q_{\mathrm{eff}} \approx 0$. Learning-rate sweeps show that flat/Haar Adam can improve finite-time risk, but this does not imply a spectral exponent improvement. This motivates reporting both best risk and q_{eff}.

5.3 Visible-profile interpolation

A controlled stochastic experiment uses fixed preconditioners

$$P_{\theta,i} = (\lambda_i^{\theta} + \rho)^{-1/2}.$$

For $a = 1.5, b = 1.4$, tuned final risk decreases monotonically as θ increases, and the fitted risk slopes steepen accordingly.

5.4 Compute-optimal scaling

Compute-optimal sweeps optimize the proxy $M^{1-b} + K^{1-b} + K/N$ under $C = MN$. In a saturating case ($a = 1.5, b = 1.4$), the risk exponent plateaus once θ crosses the phase boundary. In a hard-source full-range case ($a = 3, b = 1.4$), the compute exponent improves monotonically with θ, as predicted.

Table 1: Stochastic visible-profile training. Increasing visible exponent θ improves tuned final risk and steepens the fitted risk slope.

θ	best LR	final risk	risk slope	predicted q_{eff}
0.00	0.03	0.3177	-0.330	0.000
0.25	0.03	0.2245	-0.399	0.125
0.50	0.03	0.1330	-0.509	0.250
0.75	0.03	0.0534	-0.724	0.375
1.00	0.03	0.0074	-1.252	0.500

Table 2: Compute-optimal hard-source run with $a = 3, b = 1.4$. The risk exponent improves monotonically with visible spectral information.

θ	$\alpha = a(1-\theta/2)$	observed risk exp.	predicted risk exp.	phase
0.00	3.000	-0.102	-0.100	time-limited
0.25	2.625	-0.113	-0.110	time-limited
0.50	2.250	-0.128	-0.123	time-limited
0.75	1.875	-0.148	-0.139	time-limited
1.00	1.500	-0.167	-0.160	time-limited

5.5 AdamW decay schedules

Weight-decay schedule experiments confirm Theorem 4. For $\delta(C) = C^{-s}$, the learned-mode exponent tracks s/α until the no-decay law is recovered. Fixed weight decay ($s = 0$) produces an asymptotic floor.

5.6 Random-feature visibility

Finally, random-feature/sketch experiments test the bridge beyond diagonal coordinates. In aligned and band-limited features, coordinatewise Adam/RMSProp/AdamW have $q_{\text{eff}} \approx 0.48$–$0.49$. In Haar and global Gaussian-sketch features, they have $q_{\text{eff}} \approx 0$, while a spectral oracle remains at $q_{\text{eff}} = 1/2$. A wide learning-rate run confirms that, when the spectrum is hidden from coordinatewise second moments, the spectral oracle can achieve lower tuned final risk.

6 Related work

Scaling laws for neural systems were systematized in empirical work on language models and compute-optimal training (Kaplan et al., 2020; Hoffmann et al., 2022). The closest theoretical baseline is the linear-regression scaling law of Lin et al. (2024), which derives sharp model/data/compute tradeoffs for one-pass SGD with sketched features. Subsequent work has extended related linear-regression scaling laws to data reuse (Lin et al., 2025). Our work is complementary: we keep the linear/Gaussian setting but vary the optimizer and show how optimizer choice changes exponents when the optimizer sees spectral information.

Adaptive optimization includes AdaGrad (Duchi et al., 2011), RMSProp (Tieleman and Hinton, 2012), Adam (Kingma and Ba, 2015), and AdamW (Loshchilov and Hutter, 2019). Much theory studies convergence or implicit bias of these methods. Our focus is different: we ask when coordinatewise second-moment preconditioning reshapes the covariance spectrum enough to change scaling exponents. Recent empirical work has also argued that scaling exponents can be optimizer-dependent (Ramani and Jain, 2026); our theory gives a concrete spectral mechanism and diagnostic.

Random features and sketched linear models are classical tools for studying high-dimensional learning (Rahimi and Recht, 2007; Bach, 2017). Our results show that the geometry of a feature map controls whether coordinatewise adaptivity is spectral or merely scalar.

Table 3: Random-feature/sketch wide learning-rate run. Global mixing hides spectral information from coordinatewise Adam/RMSProp, while the spectral oracle retains $q_{\mathrm{eff}} = 1/2$.

case	optimizer	best LR	final risk	q_{eff}
Gaussian sketch	Adam	0.01	8.35×10^{-4}	-0.028
Gaussian sketch	RMSProp	0.01	1.52×10^{-3}	-0.032
Gaussian sketch	Spectral oracle	0.10	1.45×10^{-4}	0.500
Haar	AdamW	0.003	7.84×10^{-4}	-0.023
Haar	RMSProp	0.003	9.50×10^{-4}	-0.027
Haar	Spectral oracle	0.03	1.16×10^{-4}	0.500

7 LIMITATIONS

The results are proved in Gaussian linear regression and related feature-coordinate models. Neural networks can have non-Gaussian gradients, nonstationary representations, and layerwise structure. Our theory suggests diagnostics—estimate second-moment spectra and q_{eff}—but does not by itself prove neural-network scaling laws. The experiments also show that finite-time risk improvements can occur without spectral exponent improvements, so empirical optimizer comparisons must distinguish constants from asymptotic exponents.

8 CONCLUSION

Adaptive optimizers do not automatically change scaling laws. They change scaling exponents when their coordinatewise second moments make the covariance spectrum visible. The visible exponent θ determines $q_{\mathrm{eff}} = \theta/2$, which determines the learned-mode count, risk law, compute-optimal allocation, and AdamW decay schedule. This gives a mechanistic answer to when optimizer choice should enter scaling-law forecasts.

A ADDITIONAL PROOFS

A.1 SOURCE STABILITY

If $P = f(\Sigma)$ is spectral, then $\widetilde{\Sigma} = P^{1/2}\Sigma P^{1/2} = \Sigma f(\Sigma)$ and $u^\star = P^{-1/2}w^\star$. For each eigenvector v_i,

$$\widetilde{s}_i = \lambda_i f(\lambda_i)\left(f(\lambda_i)^{-1/2}\langle v_i, w^\star\rangle\right)^2 = s_i.$$

For invariant band decompositions, total source mass is preserved blockwise:

$$(P_\ell^{-1/2}w_\ell^\star)^\top(P_\ell^{1/2}\Sigma_\ell P_\ell^{1/2})(P_\ell^{-1/2}w_\ell^\star) = (w_\ell^\star)^\top\Sigma_\ell w_\ell^\star.$$

This is enough for the band-averaged bias proof because filters are constant up to factors inside comparable-eigenvalue bands.

A.2 COMPUTE-OPTIMAL DERIVATION

For fixed N, minimize $K^{1-b} + K/N$ under $K \le N^{1/\alpha}$. The unconstrained optimum satisfies $K^{-b} \asymp N^{-1}$, i.e. $K \asymp N^{1/b}$. If $b \le \alpha$, this exceeds the time cap, so $K \asymp N^{1/\alpha}$; otherwise $K \asymp N^{1/b}$. Combining with $M = C/N$ and balancing M^{1-b} against the optimized N term yields Theorem 3.

A.3 DIAGNOSTIC CONSISTENCY

Suppose on a fitting window $i \in [L, U]$,

$$\lambda_i = c_\lambda i^{-a} e^{\varepsilon_i}, \qquad \mu_i = c_\mu i^{-\alpha} e^{\delta_i},$$

with $|\varepsilon_i|, |\delta_i| \le \eta$. Least-squares log–log slope estimates satisfy

$$|\widehat{a} - a| + |\widehat{\alpha} - \alpha| \lesssim \eta / \log(U/L).$$

Thus $\widehat{q} = 1 - \widehat{\alpha}/\widehat{a}$ is consistent whenever a is bounded away from zero and the fitting window has nontrivial logarithmic width. This justifies reporting q_{eff} as the primary mechanism diagnostic.

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